Credits: 4

Prerequisites: CS-UH 1001, Introduction to Computer Science, CS-UH 1002, Discrete Mathematics

Corequisites: CS-UH 1002, Discrete Mathematics

Faculty Details	**Professor**	**Teaching Assistant**
Name	Mohamad Kassab	Khalid Mengal
Email	m.kassab@nyu.edu	kqm1@nyu.edu
Office Hours	Wednesday 2:00 PM – 3:00 PM Location: C1-112	Monday 4:00-6:00 pm (on zoom) Friday 4:00-6:00 pm (in person, Location: A2-186B) For both online and in-person office-hours a slot must be booked using the following link: https://calendly.com/kqm1/office-hours

Course Details	**Day/Time**	**Location**
Lecture	Tu, TH 11:20 AM - 12:35 PM	West Administration Room 005
Recitation	TBD	TBD
Mid Term Exam	17/10/2023	West Administration Room 005
Final Exam	TBA (see Albert)	TBA (see Albert)

This course counts toward the following NYUAD degree requirements:

Majors > Computer Science > Computer Science Required Course

Course Description

Organizing and managing large quantities of data using computer programs is increasingly essential to all scientific and engineering disciplines. This course teaches students the principles of data organization in a computer, and how to work efficiently with large quantities of data. Students learn how to design data structures for representing information in computer memory, emphasizing abstract

data types and their implementation, and designing algorithms using these representations. Topics include arrays recursion, asymptotic analysis of algorithms, lists, stacks, queues, trees, hashing, priority queues, dictionaries, graph data structures etc. This course is taught using the C++ programming language.
In addition to the 2 theory lectures, this course includes a lab component of 1:15h weekly. Lab sessions are dedicated to the practice of C++ and implementation of data structure concepts covered in the theory lectures. Lab questions and problems are designed to effectively help students in the design and implementation of computerized solutions to real problems using appropriate data structures.

Course Learning Outcomes and Link to Program Learning Outcomes (PLOs)

Students who successfully complete this course will be able to:	**CLO Level of Contribution**	**Linked to Computer Science Major PLOs[1]**
1. Design appropriate data structures for solving a given problem.	High	PLO 1
2. Analyze and compare the performance of an algorithm with different data structures.	High	PLO 1
3. Demonstrate proficiency in practical usage of standard data structures.	High	PLO 4
4. Demonstrate proficiency in programming (in C++).	High	PLO 3
5. Design, develop and implement efficient and optimized programs to solve computing problems.	High	PLO 1, PLO 3, PLO 4

Teaching and Learning Methods

There are four main teaching and learning methods employed in this course:

1. Theory lectures: to present the data structure concepts and discuss their implementations in C++
2. Class discussions: During each lecture, there will be numerous questions posed by the instructor to help students engage in the topics and to promote discussion.
3. Programming assignments: There will be several homework assignments which will sharpen the students' ability to think about problems, provide implemented solutions, and analyze their algorithms.
4. Labs: The lab sessions will include supervised programming questions that will help students to gain an in-depth understanding of the various data structures by implementing them. Students will also gain expertise in C++ and learn about the debugging process. Some lab quizzes will also be given to assess the students' practice abilities.

Assignments

- There will be 3 assignments. All assignments are due at 11:55pm on the due date marked on the course schedule. Submission will be done through NYU LMS (Brightspace).
- For late submissions, a penalty of 5% per day will be deducted for assignments submitted late, up to a maximum of 3 days. Beyond this period, a zero grade will be assigned for the

[1] See Appendix 1

assignment. No submission will be accepted 3 days after the due date.

- Students must follow the code of conduct provided with every assignment and adhere to the standards of academic integrity at NYUAD. Violations will be reported to the Vice Provost for Academic Affairs.
- You should not use ChatGPT or other AI tools to generate the code of your assignments.

Required Textbook

- Data Structures and Algorithms in C++ by Goodrich, Tamassia, Mount, 2nd Edition, Publisher: Wiley, Publication date: February 22, 2011.

Additional References

- Data Structures and Algorithm Analysis in C++ by Mark Allen Weiss, 4th Edition, Publisher: Pearson, Publication date: June 23, 2013.
- Data Structures Using C and C++ (2nd Edition). Yedidyah Langsam, Moshe J. Augenstein, Aaron M. Tenenbaum. ISBN: 978-0130369970. Pearson. (e-book not available)
- Algorithms in C++, Parts 1-4: Fundamentals, Data Structure, Sorting, Searching, 3rd Ed, July 1998. ISBN: 978-0201350883. (e-book not available)

Graded Activities

Activity Detail	Grade Percentage	Submission Date/Week	Linked to Course Learning Outcome(s)
Quizzes	15%	4 quizzes (see schedule)	1, 3
Assignments	20%	3 Assignments (see schedule)	1, 2, 3, 4, 5
Lab	10%	Ongoing	1, 2, 4, 5
Midterm Exam	25%	Week 7	3, 4
Final Exam	30%	Exam week	3, 4

Academic Policies

Attendance: Attendance for this course is mandatory. Every absence needs to be agreed with the professor prior to class. Students who, in the judgment of the instructor, have not substantially met the requirements of the course or who have been excessively absent may receive a grade reduction, including the possibility of an F. The minimum attendance rate required for this course by the instructor is 70%.

"Attendance is expected in all classes. Although the administration of NYUAD does not supervise the attendance of classes, it supports the standards established by instructors. Students who, in the judgment of the instructor have not substantially met the requirements of the course or who have been excessively absent may receive a grade reduction, including the possibility of an F, and/or may be considered to have withdrawn unofficially."

https://intranet.nyuad.nyu.edu/faculty-resources/academics/global-education-programs/j-term/faculty-j-handbook/j-term-policies/

Class Participation: Students are expected to be active participants in class discussions and activities.

Grade Distribution: Students need to obtain a grade of C or better to count the course towards their intended degree for required courses or economics electives. Course percentages will be translated into letter grades based on these intervals:

A	A-	B+	B	B-	C+	C	C-	D	F
[95,100]	[90,95)	[87,90)	[83,87)	[80,83)	[77,80)	[73,77)	[70,73)	[63,70)	[0,63)

Integrity: At NYU Abu Dhabi, a commitment to excellence, fairness, honesty, and respect within and outside the classroom is essential to maintaining the integrity of our community. By accepting membership in this community, students, faculty, and staff take responsibility for demonstrating these values in their own conduct and for recognizing and supporting these values in others. In turn, these values create a campus climate that encourages the free exchange of ideas, promotes scholarly excellence through active and creative thought, and allows community members to achieve and be recognized for achieving their highest potential.

Students should be aware that engaging in behaviors that violate the standards of academic integrity will be subject to review and may face the imposition of penalties in accordance with the procedures set out in the NYUAD policy: https://students.nyuad.nyu.edu/campus-life/student-policies/community-standards-policies/academic-integrity/

Moses Center for Student Accessibility

New York University is committed to providing equal educational opportunity and participation for students with disabilities. CSD works with NYU students to determine appropriate and reasonable accommodations that support equal access to a world-class education. Confidentiality is of the utmost importance. Disability-related information is never disclosed without student permission. Find further information at:

https://www.nyu.edu/students/communities-and-groups/students-with-disabilities.html

Contact: mosescsd@nyu.edu

Course policy on Chat-GPT and AI tools: It is important that the written work required by the course is yours. You should not use ChatGPT or other AI tools for any purpose other than idea generation. When you use any of these tools, you must include a note describing how you used them with the assignment.

Course Topics and Schedule

Below is a provisional schedule of the topics to be covered in this course. Dates are subject to confirmation and may change. Please make sure to go over the readings before coming to class.

Week	Session	Topic	Reading	Assignments and Quizzes
1	Tuesday, 29/9	Class Introduction Basics of C++	Chapter 1	
	Thursday, 31/8	Reference, arrays, classes	Chapter 1	
2	Tuesday, 5 /9	Object Oriented Design: Inheritance, Templates, Exceptions	Chapter 2	
	Thursday, 7/9	Arrays, linked lists	Chapter 3	
3	Tuesday, 12/9	Linked list	Chapter 3	**A1, Released**
	Thursday, 14/9	Recursion	Chapter 4	**Quiz 1**
4	Tuesday, 19/9	Algorithm analysis tools: Asymptotic Notation	Chapter 4	
	Thursday, 21/9	Stacks	Chapter 5	
5	Tuesday, 26/9	Queues & Deques	Chapter 5	
	Thursday, 28/9	No class		
6	Tuesday, 3/10	Vectors	Chapter 6	
	Thursday, 5/10	Iterators, Lists	Chapter 6	**A1, Due**
7	Tuesday, 10/10	Sequences & Bubble Sort	Chapter 6	**Quiz 2**
	Thursday 12/10	Midterm Review		
8	Tuesday, 17/10	Mid-term exam	N/A	(in class)
	Thursday, 19/10	No class		

Week	Session	Topic	Reading	Assignments and Quizzes
9	Tuesday, 24/10	No class		
	Thursday, 26/10	General Trees	Chapter 7	**A2, Released**
10	Tuesday, 31/10	Tree Traversal, Binary Trees	Chapter 7	
	Thursday, 2/11	Binary Trees	Chapter 7	
11	Tuesday, 7/11	Priority Queues	Chapter 8	
	Thursday, 9/11	Heaps	Chapter 8	
12	Tuesday, 14/11	Maps, Hash Tables	Chapter 9	**Quiz 3**
	Thursday, 16/11	Hash Function, Skip List, Ordered Maps, Dictionaries	Chapter 9	**A2, Due (17/11)**
13	Tuesday, 21/11	Binary Search Trees	Chapter 10	**A3, Released**
	Thursday, 23/11	AVL Trees, Multi-Way Search Trees	Chapter 10	
14	Tuesday, 28/11	(2,4) Trees, Red-Black Trees	Chapter 10	
	Thursday, 30/11	No class		
15	Tuesday, 5/12	Merge Sort	Chapter 11	**Quiz 4**
	Thursday, 7/12	Select Sort	Chapter 11	
16	Tuesday, 12/12	Additional topics in Data structures	TBD	**A3, Due**
	Thursday, 14/12	Course review	-	
Final Exam : Saturday 16-12-2023 4:45PM - 6:45PM				

Appendix 1

Computer Science Major Program Learning Outcomes (PLOs)

PLO 1 Analyze a problem, and identify, define, and verify the appropriate computational tools required to solve it (Knowledge, Skill, Role in Context, Self-development).

PLO 2 Apply up-to-date computational tools necessary in a variety of computing practices (Knowledge, Skill, Autonomy & Responsibility, Self-development).

PLO 3 Implement algorithms as programs using modern computer languages (Knowledge, Skill).

PLO 4 Apply their mathematical knowledge to solve computational problems (Knowledge, Skill, Autonomy & Responsibility, Self-development).

PLO 5 Communicate computer science knowledge both orally and in writing (Skill, Autonomy & Responsibility, Role in Context).

PLO 6 Collaborate in teams (Skill, Autonomy & Responsibility, Role in Context).